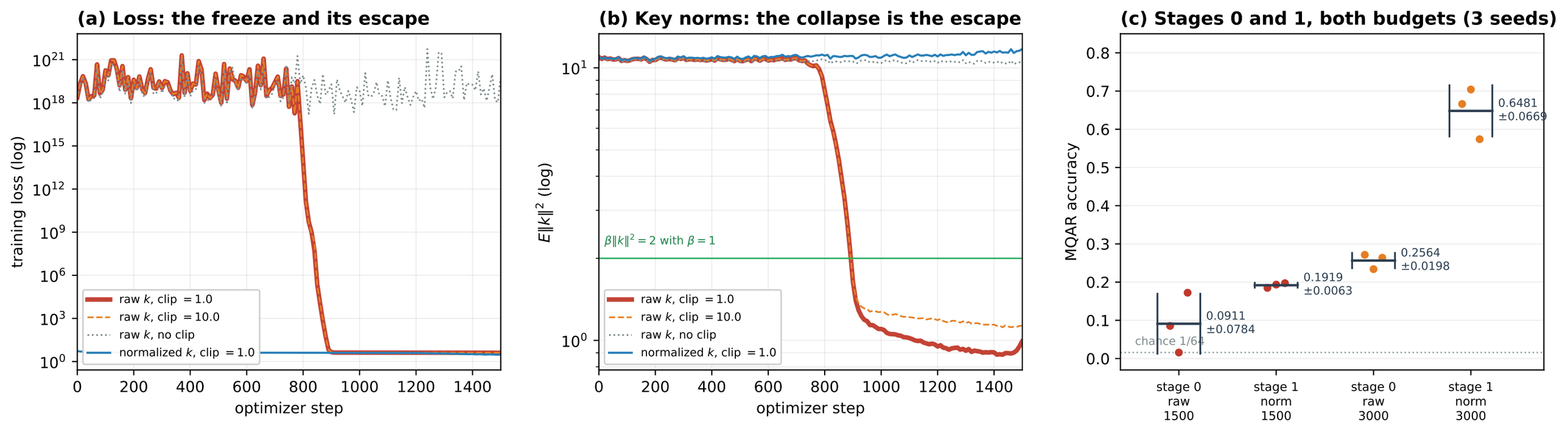
(a) Loss: the freeze and its escape
training loss (log)
optimizer step
raw k, clip = 1.0
raw k, clip = 10.0
raw k, no clip
normalized k, clip = 1.0
(b) Key norms: the collapse is the escape
E‖k‖² (log)
optimizer step
β‖k‖² = 2 with β = 1
raw k, clip = 1.0
raw k, clip = 10.0
raw k, no clip
normalized k, clip = 1.0
(c) Stages 0 and 1, both budgets (3 seeds)
MQAR accuracy
chance 1/64
0.0911
±0.0784
0.1919
±0.0063
0.2564
±0.0198
0.6481
±0.0669
stage 0
raw
1500
stage 1
norm
1500
stage 0
raw
3000
stage 1
norm
3000